



上海交通大学

SHANGHAI JIAO TONG UNIVERSITY

研究生学位论文开题报告

Graduate Thesis/Dissertation Proposal

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专业 Major	数字文创与管理
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填 报 说 明

Instruction

1. 校本部研究生的开题报告应通过[数字交大](#)在线提交申请，填写本表并上传系统。特殊情况下经研究生院事先同意，可不上传系统，并使用《上海交通大学研究生论文开题评审表》完成评审。

The application for thesis/dissertation proposal should be submitted online through [My SJTU](#). The student shall fill this form and upload it in the system. Under special circumstance, this form does not need to be uploaded and the review can be proceeded with the review form with prior consent from the graduate school.

2. 开题报告为 A4 大小，于左侧装订成册。各栏空格不够时，请自行加页。考核前提前一周送交导师、评审专家审阅。

This form should be printed with A4 papers and bound together on the left. If the space left is not enough, please feel free to add extra pages. The print version shall be sent to the supervisor, and the review committee members for review at least one week before the oral presentation.

3. 博士生导师可以根据博士生学位论文选题情况自行确定是否进行开题查新，博士学位论文开题查新报告应由查新工作站提供。

The supervisor should decide, based on the proposed topics, whether a novelty assessment report is needed or not, which should be conducted by an authorized novelty assessment department.

4. 开题报告通过后，定稿版开题报告由研究生、导师各存档一份，无需上传系统。

Upon passing the proposal, the final version of this report shall be archived by the graduate student and his/her supervisors for future reference.

5. 同等学力研究生开题答辩采用会议形式，硕士邀请至少 3 名相关学科/专业领域具有硕士研究生指导资格的专家。博士邀请 5 名相关学科/专业领域具有博士研究生指导资格的专家。

The capstone presentation adopts a conference format, and at least three experts with master's degree guidance qualifications in relevant disciplines and professional fields are invited for the master's degree. And five experts with doctoral guidance qualifications in relevant disciplines/professional fields are invited for doctoral guidance.

论文题目 Proposed Title	全链路 AI 辅助的数字油画 APP 交互设计研究 Research on Interaction Design of Full-Link AI-Assisted Digital Oil Painting APP
研究课题来源 Source of Research Project	请在合适选项前画√ Please select proper options by “√” . ☒ 国家自然科学基金课题 NSFC Research Grants ☐ 国家社会科学基金 National Social Science Fund of China ☐ 国家重大科研专项 National Key Research Projects ☐ 其它纵向科研课题 Other Governmental Research Grants ☐ 企业横向课题 R&D Projects from Industry ☐ 自拟课题 Self-proposed Project ☐ 其它 Other _____

- 1、请综述课题国内外研究进展、现状、挑战与意义，可分节描述。博士生不少于 10,000 汉字，硕士生不少于 5,000 汉字。请在文中标注参考文献。 Please review the frontier, current status, challenges and significance of the research topic. The citations should be marked in the context and listed in order at the end of this section. No less than 8,000 words for doctoral students and 4,000 words for master students if written in English.

本人承诺，开题报告第一部分文献综述中文字数 6235，英文字数 3739。
若与事实不符，本人愿承担相应后果。

1. 文献综述

本文课题标题为“全链路 AI 辅助的数字油画 APP 交互设计研究”，故本综述将针对油画绘画类型的研究价值、全链路 AIGC 辅助数字油画创作的交互逻辑优化、油画画笔库与画布库质量不佳与数量稀少的缺乏工具数据库设计的痛点、“绘画前的 AI 灵感激发 - 绘画中的丰富笔刷库与画布库 - 绘画后的油画草图智能优化 - 完成后的社区分享”的闭环交互形式创新、AIGC 赋能数字油画入门与艺术普及的体验问题等领域的多个方面，对国内外研究进展、应用现状、核心挑战与未来机遇等进行论述，用于论证本课题在全链路 AIGC 与数字油画的深度融合、面向真实绘画体验的交互设计创新（丰富的画笔库与画布库以及效率化操作设计）、以及 AIGC 驱动的数字油画体验与社区生态构建（如降低创作门槛与作品传播价值）方面的独创性、实用性及深远意义。

1.1 油画绘画类型的独特性与数字油画 APP 研究价值

1.1.1 油画画笔种类

油画画笔和画布作为油画创作中最核心的物质工具，其类型与性能的丰富程度直接决定了笔触语言和画面肌理的表达能力。近年来，国内外艺术材料学、博物馆修复学以及数字绘画研究，对画笔与画布的分类体系和性能差异进行了较为系统的梳理与量化分析，同时也从侧面揭示出当前数字油画 APP 在画笔库与画布库设计上的明显不足。

从画笔维度看，国外材料学与艺术教育机构通常从“毛料—笔形—尺寸—用途”四个维度对油画笔进行系统划分。英国 Jackson's Art 画材研究团队在 2021 年发布的画笔指南中，将油画常用毛料概括为猪鬃、貂毛、獾毛、松鼠毛与多种合成纤维，并通过教学测试指出：硬挺的猪鬃笔适合厚涂与明显纹理效果，柔软的貂毛与

獾毛更适用于细腻过渡和精细刻画，软毛毛笔则多用于罩染与边缘柔化。

Daler-Rowney 公司在 2024 年更新的画笔手册中进一步区分了天然鬃毛与人造纤维在弹性、耐溶剂性上的差异，强调不同媒介（油画、丙烯）的用笔选择应当分别配置，而非单一笔种通用。在笔形方面，英国与南非国家的课堂绘画教学中，在长期课堂实践基础上，将油画笔归纳为圆头、长平头、短平头（bright）、榛形（filbert）、扇形、排笔、线描笔等，并通过示范课程证明，同一尺幅作品中通常需要 4-6 种不同形状与规格的画笔搭配，才能兼顾大形块概括、厚涂肌理与细节刻画。

国内教学与画材科普在引介上述分类体系的基础上，加入了本土使用经验。2024 年台湾 StyleArt（造型美术）画材团队在台北面向高校学生的教学中，将油画笔按“软硬程度 × 用途”划分为猪鬃类硬笔、貂毛与牛耳毛中软笔、化纤综合笔三大类，分别对应厚涂铺色、人物细节与特效肌理，并指出职业画家通常会配备十余支不同材质和形状的画笔，以应对写实、厚涂、薄涂等多种技法场景。通过这些研究可以看到，油画画笔在毛料、笔形与型号组合上的高度多样性，使得现实工作室中常用的工具组合远超一般数字软件中“几款通用画笔”的配置规模。

1.1.2 油画画布种类

关于油画画布的种类，博物馆修复与材料科学研究揭示了画布在纤维材质、织造纹理与底层涂布工艺上的复杂性。英国南安普顿大学和泰特美术馆合作的“Canvas Complexity”项目在 2010 年以后对多家画布厂商样本进行分析，发现当前艺术家常用画布至少包含亚麻、棉、混纺与聚酯等多种纤维类型，不同纤维在强度、伸缩性和吸油性上存在显著差异，直接影响画面耐久性与笔触反馈。荷兰 Royal Talens 公司在其 2023 年画布产品手册中，将专业油画布划分为亚麻系列和棉布系列，并通过出厂检测给出不同系列在织物克重、纹理粗细与底料涂层（丙烯基底、油性基底）方面的参数说明，强调亚麻布在尺寸稳定性与长久性方面优于棉布，而粗纹与细纹画布分别适配厚涂与细腻写实风格。

中国画材科普网站与教程文章在 2018-2024 年间陆续将手绘油画布分为亚麻、纯棉、棉麻混纺、化纤及仿宣纸纹理等多个品类。这些不同的品类都会直观影响油画的画笔落在画布上的直观表现与绘画效果。美国阿拉巴马艺术用品公司在 2025 年发布的“Canvas vs. Paper”教学文章中，通过示范课程比较大尺幅油画在纸张与绷画布上的表现，指出绷好的棉布与亚麻画布在弹性反馈、厚涂承载能力及长久性方

面明显优于纸质载体，并强调“画面是否能在几十年后保持稳定，很大程度取决于画布基础结构与底料处理”。

由此可见，画笔与画布的类型与知识对于油画绘画的重要性与油画绘画过程表达的影响。

1.1.3 油画画笔库与画布库知识对艺术的作用

画笔与画布的多样性研究已经成为艺术史与技法研究的重要支撑。Gombrich 在《艺术的故事》中通过对 19 世纪以来绘画语言嬗变的梳理指出，金属箍套平头画笔的普及改变了笔触形态，为印象派大色块分割和显露笔触的风格提供了技术条件。国内部分艺术院校的课程方案统计也显示，画笔与画布材料的理论与实践教学在油画技法课程中占据较高比重，部分院校将其学时比例提高到接近四分之一，表明学界普遍认可“工具知识”在技法教育与审美训练中的基础性地位

1.1.4 油画画笔库与画布库对于创作体验的影响

在实践层面，多项面向职业画家与学生的教学与调研结果表明，画笔与画布选择直接影响创作体验、作品质感和保存安全。南非画家 Malcolm Dewey 在 2019 年的教学博客与工作坊中，通过对比廉价合成纤维笔与高品质猪鬃、貂毛笔在厚涂、干擦等技法中的表现，指出优质画笔在保持笔形、承载颜料和减少掉毛方面显著优于廉价替代品，并总结“好画笔会让画面更容易达成预期效果，也会显著提升创作过程中的愉悦感”。阿拉巴马艺术用品公司则在 2025 年的公开课程中，通过示范不同规格棉布与亚麻绷画布在大面积涂布、分层罩染和厚涂刮擦中的反应，向学生强调“错误的画布选择会带来开裂、下陷或变形等长期风险”，将材质选择纳入基础教学的重要环节。这些一线教学与工作室经验，从另一个侧面印证了画笔与画布研究的实践价值，它不仅关乎艺术史考证，更关乎当代创作的手感与作品寿命。

1.1.5 数字油画 APP 的画笔库与画布库痛点问题

与上述现实工具生态相比，当前数字油画与数字绘画软件在“画笔库—画布库”设计上存在明显结构性差距。一方面，多数通用数字绘画软件虽然声称提供“海量画笔”，但这些预设更多按照效果与风格类别组织，而不是严格对应传统油画中的毛料、笔形与型号体系。例如，Procreate 官方手册在 2020 年以后提供了 18 组核心笔刷集与若干扩展包，但主要以“素描类、墨线类、绘画类”等功能性标签进行划分，而非按照“猪鬃圆头 8 号”“亚麻粗纹厚涂笔”等传统材料逻辑来组织。Adobe Fresco 则依托 Creative Cloud 生态提供数以千计的像素笔刷和基于物理引擎的“Live Brushes”，其官方文档与媒体报道强调这些笔刷能够模拟水彩晕染和油画厚涂的流动效果，但同样是建立在统一算法引擎上的“效果分类”，而非一一映射现实画笔与画布的具体组合。2020 年以来，多篇针对 ArtRage、Corel Painter 等软件的中文评测文章在北京、上海等地的数字绘画培训中被用作教材，它们普遍肯定 ArtRage 在油画厚涂和调色刀效果上的优势，但也指出工具种类主要集中在“油刷、铅笔、喷枪、调色刀、滚筒”等有限几类，难以覆盖现实中多种毛料与特型画笔所产生的微妙差异。

国内外关于油画画笔与画布的材料学、教学与修复研究，已经充分证明了现实工具体系在材质、形态与工艺组合上的高度多样性及其在艺术史与教学中的关键价值；而现有数字油画与数字绘画软件在画笔库与画布库设计上，无论在品类覆盖、参数细化还是肌理生成机理上，都与现实油画工具体系存在显著落差。这一差距不仅制约了数字作品在笔触质感与材料语言上的表达，也为后续“全链路 AI 辅助数字油画 APP”在工具库设计与交互优化方面提供了清晰的研究靶点与改进方向。

1.2 现有数字油画 APP 交互设计的研究现状与缺陷

1.2.1 国内外数字油画 APP 交互设计研究现状

从产业侧看，WiseGuyReports (2023) 基于多家软件厂商财报与访谈，对全球数字绘画市场进行测算，指出 2023 年数字绘画市场规模约为 71.2 亿美元，并预测到 2032 年将增长到 225.3 亿美元，主要增量来自移动端绘画软件与在线创作平台，这意味着数字油画 APP 已成为在线创作与美育的重要基础设施之一。

在具体功能层面，国内开发实践往往以单一绘画应用为对象进行模块化拆分设计。以 App Inventor 移动绘画应用的开发实践为例，朱吉虹等 (2022) 在一篇关于“基于用户粘性提升的美育 App 交互设计研究”中，将绘画类 App 功能划分为画布、

工具栏、颜色选择器、图层管理、保存与分享等基础模块，并重点讨论界面布局与操作流程的简化，但对作品管理、社群互动、创作激励等进阶功能的交互机制着墨较少。

国外研究在笔触模拟与物理媒介建模上取得了显著突破，但大量工作以算法与渲染为中心，真实触觉反馈与学习体验往往被当作“后置问题”。这在现实中表现为，即便像 ArtRage、Corel Painter 等商业软件已经具备丰富笔刷库与油画仿真模块，其核心竞争力仍集中在画面效果，而非从交互设计角度系统解决“怎么让用户感觉在画真正的油画”。而国内的数字用户 app 研究，一方面，研究者和开发者已经形成“画布+工具栏+课程内容”的基本交互共识；另一方面，无论是绘画美育平台还是绘画工具类 App，其交互设计研究仍多停留在“工具是否好用”“界面是否易学”的局部层面，对涵盖“灵感—创作—优化—分享”的全链路数字油画体验尚缺乏系统的设计和实证分析。

1.2.2 现有油画绘画 APP 交互设计的核心缺陷

近年来，多家市场研究机构在分析数字艺术软件与数字绘画软件时均指出：用户对“智能化、低门槛”的创作支持需求不断上升，而现有产品在工作流层面的智能辅助仍较薄弱。Verified Market Reports（2023）在《Digital Art Software Market》报告中，通过对专业艺术家与业余爱好者的调研发现，虽然越来越多软件集成了 AI 上色、照片转画作等功能，但这些功能多集中在图像后期或单步风格迁移，难以贯穿“灵感生成—构图草图—细节精修—成品导出”的完整流程。

在具体产品层面，Corel 公司在 Painter Essentials 8 的官方教程与宣传中，将“AI-based photo painting”“一键照片转画作”作为核心卖点之一，主要场景是用户导入照片后由系统自动生成类油画效果，再进行少量笔触修饰。这一功能路径事实上验证了报告中的判断：AI 多被嵌入为局部效果滤镜，而不是被设计为对创作全过程提供连续反馈与建议的“共创伙伴”。

面向交互设计的 AIGC 研究也给出了类似警示。李早临与汤天轶（2024）在南通大学进行的 AIGC 实时绘画实验中，基于交互装置构建“实时生成+人机共创”的绘画系统，指出当前多数 AIGC 绘画应用仍停留在“生成结果”的层面，缺乏对互动性、可控性和过程参与感的精细设计，因此很难有效嵌入专业创作或教学场景。

从用户侧看，东吴证券等机构在 2023 年发布的 AIGC 细分赛道研究中，对国内 AI 绘画用户进行问卷调查，结果显示约 64% 的用户为学生群体，美术与设计专业人群占比不足 1/3，大量使用场景是兴趣尝试与社交分享。研究者据此推断，目前 AI 绘画工具更像“生成玩具”而非“创作助手”，尚未在专业创作流程中形成高强度黏性，这与智能功能缺乏对从草图、色稿到终稿的全链路支持密切相关。

现有数字油画 APP 在功能结构上存在两类断裂：一是灵感与构图阶段缺乏针对油画语言的智能引导，如参考作品推荐、结构拆解与用色建议等；二是细节精修与作品评估环节多依赖用户个人经验，缺少基于经典油画技法的 AI 审阅与局部优化工具。这种“有 AI 特效、无 AI 流程”的状态，使得 AI 辅助难以真正降低油画学习与创作门槛，并且丧失了绘画创作的乐趣，ai 并没有作为创作的辅助工具而是只关注结果的“捷径”。

1.3 AIGC 融入数字油画 APP 交互设计的依据、现状与适配性

1.3.1 AIGC 融入数字油画的核心依据

从生成能力看，扩散模型已经在艺术图像合成与风格控制上被系统证明具备高质量输出。Wang 等人 2024 年在香港理工大学主导的扩散艺术创作综述中，梳理了近百篇相关论文，指出 Stable Diffusion 等潜空间扩散模型在油画、版画、国画等多种艺术风格上都能稳定生成结构完整、风格一致的图像，并在人类主观审美和自动评价指标上明显优于早期 GAN 与传统风格迁移方法。Rombach 等 2021 年在德国海德堡大学的实验则表明，潜空间扩散模型在同一框架下即可完成文本生成图像、风格化与超分辨等多任务，证明这一技术路线具备“一套模型覆盖多环节”的潜力。

扩散模型及其变体在灵感生成、过程重建、风格化转换与高清化上已经形成相对成熟的算法谱系，为数字油画 APP 补齐“灵感—创作—精修—高清导出”中长期缺位的功能模块提供了坚实基础。

而从学习者视角看，多项教育实证研究证明 AIGC 在绘画学习中的“减负”与“增效”作用。Bian 等人 2025 年在中国某小学开展的随机对照实验中，将 78 名五年级学生分为 AI 辅助组与对照组，结果显示，在将 Stable Diffusion 生成的图像

嵌入美术课堂后, AI 组学生的课堂投入度与自我效能显著提升, 认知负荷显著下降。这说明对于缺乏专业基础的学习者来说, AIGC 生成的参考图与示范图能有效缓解“无从下手”的焦虑。

从更宏观的教育视角, Rong 等人 2025 年对高等教育中生成式 AI 艺术应用的系统综述指出, 生成式 AI 工具在艺术教育中的典型作用主要体现在三类场景: 一是为学生提供大量风格各异的视觉参考, 支持快速发散构思; 二是在创作中提供构图、色彩和细节调整的即时反馈; 三是在课外作业中充当“辅助指导”, 降低反复试错的成本。Sun 的研究与这一综述共同表明: 只要合理设计交互, AIGC 可以从“好玩工具”转变为“降低门槛、提升坚持度的教学基础设施”。

在数字媒体艺术创作侧, Li 2024 年关于 AIGC 在数字媒体艺术创作中实践的研究指出, 当前艺术家和学生在使用 AIGC 时, 最看重的正是“快速生成参考草图”“风格迁移”和“局部修补”三类功能, 这与前述教育研究中的“构思—调整—完稿”三阶段高度对应。

综上, 结合这些实证结果, 可以推论, 如果数字油画 APP 能将 AIGC 模块有机融入, 从灵感推荐、草图辅助到成品精修, 就能正面回应新手用户“没人指导、效率太低”的核心痛点, 同时减轻一线教师在技法示范与个别辅导上的压力。

1.3.2 AIGC 融入数字油画的局部突破与全链路空白

从学术研究版图看, AIGC 在绘画与视觉艺术中的应用已经非常活跃, 但大多集中在单点技术或单一环节。Wang 等人 2024 年的扩散艺术创作综述显示, 近几年相关论文主要围绕“更高质量的图像生成”“更精细的风格控制”和“更鲁棒的文本对齐”展开, 对“如何把生成模型嵌入完整创作流程”“如何与交互界面、教学流程深度整合”的讨论比例明显偏低。Cetinić 与 She 等人 2022 年在 ACMTIST 综述中也指出, 现有大部分研究聚焦算法性能和审美效果, 而对人机共创过程、创作 workflow 与美术教育场景的系统化设计仍属薄弱环节。

在教育应用层面, Rong 等人 2025 年综述了高等教育中生成式 AI 艺术应用的 40 余项案例, 发现多数项目将 AIGC 用作某一教学活动中的“辅助手段”, 例如用 text-to-image 生成人物形象供学生进行再创作, 或用风格迁移工具帮助学生理解不同画派风格; 较少有研究针对“从选题、构思、草图、上色到评价与展示”的完

整学习流程进行一体化设计。Li 等人 2025 年的 AIGC 准实验课程同样主要将生成工具嵌入设计作业环节，并未进一步讨论如何将其机制固化为一套可复用的数字产品交互范式。

产业实践侧，大量商用绘画与设计软件已经引入 AIGC 功能，但更多是以“局部增强模块”的形式存在。Adobe 在 Photoshop 与 Firefly 体系中推出的 Generative Fill、Generative Expand 与 Generative Upscale，可在局部编辑、画面扩展与质量提升上显著简化操作流程，但其主要面向平面设计与图片编辑任务，并未针对油画技法学习构建专门的创造流程。

Clip Studio Paint 等绘画软件在 2023 年后陆续上线 AI 姿势生成、智能线稿优化、AI 色调调整等“智能工具”，官方文档与独立测评均强调其作用在于“加速部分操作、改善线条与色彩质量”，而非重构整套创作流程。

所以，总体来看，无论学界还是业界，当前 AIGC 与艺术创作结合的典型模式，大多停留在在“灵感阶段”通过 text-to-image 生成参考图、“创作中期”提供局部风格迁移或构图建议、“成品阶段”使用超分辨和自动润饰工具提升画面质量。而真正围绕“油画”这一特定媒介，把 AIGC 与笔触技法、层层罩染、教学节奏和作品分享一并纳入统一交互框架的全链路研究，目前在公开文献中仍基本空白。这也为本研究与数字油画 APP 的交互设计提供了清晰的“研究空位”：既要承接已有的局部技术突破，又要在产品层面完成从“功能拼盘”到“整体体验”的结构重构。

1. Literature Review

The title of this thesis is “Research on Interaction Design of Full-Link AI-Assisted Digital Oil Painting APP”. Therefore, this review addresses several aspects of the field, including the research value of oil painting typologies, the optimization of interaction logic in full-link AIGC-assisted creation, and the pain points caused by the poor quality and scarcity of oil brush and canvas libraries in current tools. It also examines the innovation of a closed-loop interaction form, characterized as “AI inspiration before painting, rich brush and canvas libraries during painting, intelligent optimization of oil sketches after painting, and community sharing upon completion.” Furthermore, it discusses the user experience issues related to AIGC-empowered digital oil painting for beginners and art popularization. By discussing domestic and international research progress, current application status, core challenges, and future opportunities, this review serves to demonstrate the originality, practicality, and profound significance of this project. Specifically, it focuses on the deep integration of full-link AIGC with digital oil painting, interaction design innovation oriented toward authentic painting experiences (through rich tool libraries and efficient operation design), and the construction of an AIGC-driven digital oil painting experience and community ecosystem (such as lowering the barrier to creation and enhancing the dissemination value of works).

1.1 The Uniqueness of Oil Painting Types and the Research Value of Digital Oil Painting Apps

1.1.1 Types of Oil Paintbrushes

Oil paintbrushes and canvases are the most fundamental physical tools in oil painting creation. The richness of their types and performance directly determines the expressive ability of brushstroke language and surface texture. In recent years, research in art materials science, museum conservation, and digital painting has systematically organized and quantitatively analyzed the classification systems and performance differences of brushes and canvases. This research indirectly reveals significant deficiencies in the design of brush and canvas libraries in current digital oil painting apps.

From the perspective of brushes, foreign institutions of materials science and art education typically classify oil brushes based on four dimensions: bristle material, brush shape, size, and usage. In a brush guide published in 2021, the research team at Jackson’s Art in the UK summarized common oil painting materials as hog bristle, sable, badger, squirrel hair, and various synthetic fibers. Through instructional testing, they pointed out that stiff hog bristle is suitable for impasto and obvious texture effects, soft sable and badger hair are

better for delicate transitions and fine detailing, while soft hair brushes are mostly used for glazing and softening edges. In a brush manual updated in 2024, Daler-Rowney further distinguished the differences in elasticity and solvent resistance between natural bristles and synthetic fibers, emphasizing that brush selection should be configured separately for different media (oil versus acrylic) rather than using a single brush type for all purposes. Regarding brush shapes, classroom painting instruction in the UK and South Africa, based on long-term practice, categorizes oil brushes into round, flat, bright, filbert, fan, and rigger, among others. Demonstration courses have proven that a single work usually requires a combination of 4 to 6 different shapes and specifications of brushes to balance large-shape generalization, impasto texture, and detailed depiction.

Domestic teaching and art material popularization have introduced these classification systems while incorporating local usage experience. In 2024, the StyleArt team in Taiwan classified oil brushes for university students into three major categories based on

“softness/hardness and usage” : hard brushes (hog bristle), medium-soft brushes (sable and ox hair), and synthetic composite brushes. These correspond respectively to thick color application, character details, and special texture effects. They pointed out that professional painters usually equip themselves with more than ten brushes of different materials and shapes to handle various technical scenarios such as realism, impasto, and thin coating. These studies show that the high diversity of oil paintbrushes in terms of materials, shapes, and model combinations makes the tool combinations commonly used in real studios far superior to the scale of "a few general brushes" configured in typical digital software.

1.1.2 Types of Oil Canvases

Regarding types of oil canvases, research in museum conservation and materials science reveals the complexity of canvases in terms of fiber material, weave texture, and priming processes. The “Canvas Complexity” project, a collaboration between the University of Southampton and Tate Modern, analyzed samples from multiple canvas manufacturers after 2010. They found that canvases commonly used by contemporary artists include at least linen, cotton, blends, and polyester fibers. Different fibers show significant differences in strength, elasticity, and oil absorption, which directly affects the durability of the painting and brushstroke feedback. In its 2023 canvas product manual, the Dutch company Royal Talens divided professional oil canvases into linen series and cotton series. Through factory testing, they provided parameter explanations for different series regarding fabric weight, texture coarseness, and priming layers (acrylic base versus oil base), emphasizing that linen is superior to cotton in dimensional stability and longevity, while coarse and fine textures are adapted to impasto and delicate realistic styles, respectively.

Between 2018 and 2024, Chinese art material websites and tutorial articles successively

classified hand-primed oil canvases into multiple categories such as linen, pure cotton, cotton-linen blends, synthetic fibers, and imitation rice paper textures. These different categories intuitively affect the visual performance and painting effect of the oil brush on the canvas. In a “Canvas vs. Paper” instructional article published in 2025, Alabama Art Supply in the USA compared the performance of large-scale oil paintings on paper and stretched canvas through demonstration courses. They pointed out that stretched cotton and linen canvases are significantly superior to paper carriers in terms of elastic feedback, impasto load-bearing capacity, and longevity, emphasizing that whether a painting can remain stable decades later largely depends on the canvas infrastructure and primer treatment.

This demonstrates the importance of knowledge regarding brush and canvas types for oil painting and their impact on the expression of the painting process.

1.1.3 The Role of Brush and Canvas Library Knowledge in Art

The study of brush and canvas diversity has become an important support for art history and technique research. In *The Story of Art*, Gombrich traced the evolution of painting language since the 19th century, pointing out that the popularization of metal-ferruled flat brushes changed the form of brushstrokes, providing technical conditions for the Impressionist style of large color block division and exposed brushstrokes. Statistics from course plans in some domestic art academies also show that theoretical and practical teaching of brush and canvas materials occupies a high proportion in oil painting technique courses. Some institutions have increased this proportion to nearly one-quarter of class hours, indicating that the academic community generally recognizes the foundational status of “tool knowledge” in technique education and aesthetic training.

1.1.4 The Impact of Brush and Canvas Libraries on the Creative Experience

On a practical level, multiple teaching and survey results targeting professional painters and students indicate that the choice of brush and canvas directly affects the creative experience, work quality, and preservation safety. In his 2019 teaching blog and workshops, South African painter Malcolm Dewey compared the performance of cheap synthetic fiber brushes with high-quality hog bristle and sable brushes in techniques like impasto and dry brush. He pointed out that quality brushes are significantly superior to cheap substitutes in maintaining shape, holding pigment, and reducing shedding, concluding that good brushes make it easier to achieve expected effects and significantly enhance the pleasure of the creative process. In open courses in 2025, Alabama Art Supply demonstrated the reactions of different specifications of cotton and linen stretched canvases during large area coating, layering glazing, and impasto scraping. They emphasized to students that incorrect canvas selection entails long-term risks such as cracking, sinking, or deformation, thereby incorporating material selection as a key part of

basic teaching. These frontline teaching and studio experiences confirm the practical value of brush and canvas research from another perspective; it concerns not only art historical verification but also the tactile feel of contemporary creation and the lifespan of the artwork.

1.1.5 Pain Points in the Brush and Canvas Libraries of Digital Oil Painting Apps

Compared to the physical tool ecosystem described above, current digital oil painting and digital painting software exhibit a significant structural gap in the design of "brush libraries" and "canvas libraries." On one hand, although most general digital painting software claims to provide "massive brushes," these presets are organized more by effect and style categories rather than strictly corresponding to the material, shape, and model systems of traditional oil painting. For example, since 2020, the official Procreate manual has offered 18 core brush sets and various expansion packs, but they are primarily classified by functional labels such as "sketching, inking, painting," rather than being organized by traditional material logic like "Hog Bristle Round No. 8" or "Linen Coarse Texture Impasto Brush." Adobe Fresco offers thousands of pixel brushes and physics-based "Live Brushes" through the Creative Cloud ecosystem. Its official documentation and media reports emphasize that these brushes can simulate the flow effects of watercolor bleeding and oil impasto, but they are still based on "effect classification" within a unified algorithmic engine, rather than mapping one-to-one to specific combinations of physical brushes and canvases. Since 2020, multiple Chinese review articles focusing on software like ArtRage and Corel Painter have been used as teaching materials in digital painting training in Beijing and Shanghai. They generally affirm ArtRage's advantages in oil impasto and palette knife effects but point out that tool types are mainly concentrated in a few limited categories like "oil brush, pencil, airbrush, palette knife, roller," making it difficult to cover the subtle differences produced by various bristles and special-shaped brushes in reality.

Domestic and international research on materials science, teaching, and conservation of oil painting brushes and canvases has fully proven the high diversity of the physical tool system in terms of material, form, and craft combinations, as well as its key value in art history and teaching. However, existing digital oil painting and digital painting software show a significant disparity in brush and canvas library design compared to the physical oil painting tool system, whether in category coverage, parameter refinement, or texture generation mechanisms. This gap not only restricts the expression of digital works in terms of brushstroke texture and material language but also provides clear research targets and improvement directions for tool library design and interaction optimization in subsequent "Full-Link AI-Assisted Digital Oil Painting APPs."

1.2 Current Research Status and Defects in the Interaction Design of Existing Digital Oil Painting Apps

1.2.1 Current Research Status of Digital Oil Painting App Interaction Design

From an industrial perspective, WiseGuyReports (2023), based on financial reports and interviews with various software vendors, estimated the global digital painting market. They noted that the market size was approximately 7.12 billion USD in 2023 and predicted it would grow to 22.53 billion USD by 2032. The main growth is expected to come from mobile painting software and online creation platforms, meaning that digital oil painting apps have become essential infrastructure for online creation and aesthetic education.

At the specific functional level, domestic development practices often focus on modular design for single painting applications. Taking the development practice of App Inventor mobile painting applications as an example, Zhu Jihong et al. (2022), in a study on "Interaction Design of Aesthetic Education Apps Based on User Stickiness Enhancement," divided painting app functions into basic modules such as canvas, toolbar, color picker, layer management, preservation, and sharing. They focused on simplifying interface layout and operation flow but placed less emphasis on the interaction mechanisms of advanced functions like work management, community interaction, and creation incentives.

Foreign research has made significant breakthroughs in brushstroke simulation and physical media modeling, but much of this work centers on algorithms and rendering, often treating real tactile feedback and the learning experience as secondary issues. In reality, this means that even though commercial software like ArtRage and Corel Painter already possess rich brush libraries and oil painting simulation modules, their core competitiveness remains focused on visual effects rather than systematically solving "how to make the user feel like they are painting a real oil painting" from an interaction design perspective. Meanwhile, domestic research on digital user apps shows two trends: researchers and developers have reached a basic interaction consensus of "canvas + toolbar + course content"; however, interaction design research for both painting education platforms and painting tool apps remains largely at the local level of "is the tool easy to use" or "is the interface easy to learn." There is a lack of systematic design and empirical analysis for the full-link digital oil painting experience covering "inspiration, creation, optimization, and sharing."

1.2.2 Core Defects in Existing Oil Painting App Interaction Design

In recent years, multiple market research agencies analyzing digital art software and digital painting software have pointed out that user demand for "intelligent, low-barrier" creation support is rising, yet intelligent assistance in the workflow of existing products remains weak. Verified Market Reports (2023), in its "Digital Art Software Market" report, found through surveys of professional artists and amateurs that although more software is integrating functions like AI coloring and photo-to-painting conversion, these are mostly concentrated in post-processing or single-step style transfer. They struggle to span the

complete process of "inspiration generation, composition sketching, detail refining, and finished product export."

At the product level, Corel promoted "AI-based photo painting" and "one-click photo to painting" as core selling points in the official tutorials and marketing for Painter Essentials 8. The main scenario involves users importing a photo, having the system automatically generate an oil-like effect, and then performing minor brush touch-ups. This functional path essentially verifies the judgment in the reports: AI is mostly embedded as a local effect filter rather than being designed as a "co-creation partner" that provides continuous feedback and suggestions throughout the creative process.

Interaction design research oriented toward AIGC has offered similar warnings. Li Zaolin and Tang Tianyi (2024), in a real-time AIGC painting experiment at Nantong University, built a "real-time generation + human-machine co-creation" painting system based on interactive installations. They pointed out that most current AIGC painting applications remain at the level of "generating results" and lack refined design for interactivity, controllability, and process participation, making it difficult to effectively embed them into professional creation or teaching scenarios.

From the user side, research on the AIGC niche sector released by institutions such as Soochow Securities in 2023 surveyed domestic AI painting users. The results showed that about 64% of users are students, while art and design professionals account for less than one-third. A large number of usage scenarios involve interest-based experimentation and social sharing. Researchers inferred from this that current AI painting tools are more like "generative toys" than "creative assistants" and have not yet formed high-intensity stickiness in professional creative workflows. This is closely related to the lack of full-link support from intelligent functions for stages ranging from sketches and color drafts to final drafts.

Existing digital oil painting apps suffer from two types of disconnection in their functional structure. First, there is a lack of intelligent guidance tailored to oil painting language during the inspiration and composition stages, such as reference work recommendations, structural breakdown, and color suggestions. Second, the detail refining and work evaluation stages rely mostly on users' personal experience, lacking AI review and local optimization tools based on classic oil painting techniques. This state of "having AI effects but no AI workflow" prevents AI assistance from truly lowering the threshold for oil painting learning and creation. Furthermore, it diminishes the joy of painting creation, as AI is not used as an auxiliary tool for creation but rather as a "shortcut" that focuses solely on the result.

1.3 Basis, Status, and Suitability of Integrating AIGC into Digital Oil Painting App Interaction Design

1.3.1 Core Basis for Integrating AIGC into Digital Oil Painting

From the perspective of generation capability, diffusion models have been systematically proven to possess high-quality output in artistic image synthesis and style control. In a 2024 review of diffusion art creation led by Wang et al. at Hong Kong Polytechnic University, nearly a hundred related papers were examined. The review pointed out that latent diffusion models, such as Stable Diffusion, can stably generate images with complete structures and consistent styles across various art styles, including oil painting, printmaking, and traditional Chinese painting. Moreover, these models perform significantly better than early GANs and traditional style transfer methods in terms of human subjective aesthetics and automatic evaluation metrics. Experiments by Rombach et al. at Heidelberg University in Germany in 2021 indicated that latent diffusion models could complete multiple tasks such as text-to-image generation, stylization, and super-resolution within the same framework, proving that this technical route has the potential for "one model covering multiple stages."

Diffusion models and their variants have formed a relatively mature algorithm lineage in inspiration generation, process reconstruction, stylistic transformation, and high-definition processing. This provides a solid foundation for digital oil painting apps to fill the long-missing functional modules in the "inspiration, creation, refining, and HD export" chain.

From the learner's perspective, multiple empirical educational studies prove the "burden reduction" and "efficiency enhancement" effects of AIGC in painting learning. In a randomized controlled trial conducted by Bian et al. in 2025 at a primary school in China, 78 fifth-grade students were divided into an AI-assisted group and a control group. The results showed that after embedding images generated by Stable Diffusion into the art classroom, the classroom engagement and self-efficacy of students in the AI group increased significantly, while cognitive load decreased significantly. This indicates that for learners lacking a professional foundation, reference images and demonstration images generated by AIGC can effectively alleviate the anxiety of "not knowing where to start."

From a broader educational perspective, a systematic review by Rong et al. in 2025 on generative AI art applications in higher education pointed out that the typical roles of generative AI tools in art education are mainly reflected in three scenarios: first, providing students with a large number of visual references with diverse styles to support rapid divergent thinking; second, providing immediate feedback on composition, color, and detail adjustment during creation; and third, acting as an "auxiliary guide" in extracurricular assignments to reduce the cost of repeated trial and error. Sun's research, along with this review, shows that as long as the interaction is reasonably designed, AIGC

can transform from a "fun tool" into "teaching infrastructure that lowers thresholds and improves persistence."

On the digital media art creation side, Li's 2024 research on the practice of AIGC in digital media art creation pointed out that when artists and students use AIGC, what they value most are the functions of "rapid generation of reference sketches," "style transfer," and "local in-painting." These correspond highly to the three stages of "conception, adjustment, and completion" in educational research.

In summary, combining these empirical results, it can be inferred that if digital oil painting apps can organically integrate AIGC modules—from inspiration recommendation and sketch assistance to finished product refining—they can directly address the core pain point of novice users having "no guidance and low efficiency," while also relieving the pressure on frontline teachers regarding technique demonstration and individual tutoring.

1.3.2 Local Breakthroughs and the Full-Link Gap in AIGC Integration

In the academic landscape, the application of AIGC in painting and visual arts is very active, but it is mostly concentrated on single-point technologies or single links. The diffusion art creation review by Wang et al. (2024) shows that in recent years, related papers have mainly revolved around "higher quality image generation," "finer style control," and "more robust text alignment." The proportion of discussions on "how to embed generative models into a complete creative process" and "how to deeply integrate with interaction interfaces and teaching flows" is significantly low. Cetinić and She et al. (2022) also pointed out in an ACM TIST review that most existing research focuses on algorithm performance and aesthetic effects, while the systematic design of the human-machine co-creation process, creative workflows, and art education scenarios remains a weak link.

At the educational application level, Rong et al. (2025) reviewed over 40 cases of generative AI art applications in higher education and found that most projects used AIGC as an "auxiliary means" in a specific teaching activity. Examples include using text-to-image to generate character images for students to recreate, or using style transfer tools to help students understand different art styles. Few studies have designed an integrated system for the complete learning process from "topic selection, conception, sketching, and coloring to evaluation and display." The AIGC quasi-experimental course by Li et al. (2025) similarly embedded generation tools primarily into the design assignment link, without further discussing how to solidify its mechanism into a reusable interaction paradigm for digital products.

On the industrial practice side, a large number of commercial painting and design software products have introduced AIGC functions, but mostly in the form of "local enhancement

modules." The Generative Fill, Generative Expand, and Generative Upscale features launched by Adobe in the Photoshop and Firefly system can significantly simplify operation processes in local editing, image expansion, and quality improvement. However, they are mainly oriented toward graphic design and image editing tasks and have not built a specialized creation process for oil painting technique learning.

Clip Studio Paint and other painting software have successively launched "intelligent tools" such as AI pose generation, intelligent line art optimization, and AI tone adjustment since 2023. Official documentation and independent reviews emphasize that their role lies in "accelerating some operations and improving line and color quality," rather than reconstructing the entire creative workflow.

Therefore, whether in academia or industry, the current typical mode of combining AIGC with art creation mostly stops at generating reference images via text-to-image in the "inspiration stage," providing local style transfer or composition suggestions in the "mid-creation stage," and using super-resolution and automatic retouching tools to improve image quality in the "finished stage." However, full-link research that truly centers on the specific medium of "oil painting" and incorporates AIGC with brushwork techniques, layering glazing, teaching rhythm, and work sharing into a unified interaction framework remains basically blank in open literature. This provides a clear "research void" for the interaction design of this study and the digital oil painting app: it is necessary to not only inherit existing local technical breakthroughs but also complete a structural reconstruction from a "platter of functions" to a "holistic experience" at the product level.

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2、课题研究目标、主要研究内容和拟解决的关键问题。 **Research objectives, main contents and key issues to be solved.**

2.1 课题研究目标

本课题面向全年龄段油画创作者，针对当前数字油画 APP 存在的“全链路功能断裂（灵感 - 创作 - 管理 - 分享脱节）、油画画布库与画笔库种类少并缺乏系统性整理、AIGC 辅助碎片化、社区生态与创作流程割裂”等核心问题，目标在于设计一款关注油画绘画全链路四个阶段（绘画前、绘画中、绘画完成后、分享绘画）的 AIGC 融合数字油画绘画的 app 交互设计，包含 AI 灵感发散界面、画板界面、画布库与画笔库、画廊管理界面、社区图库等界面功能。通过构建“AIGC 深度协同油画绘画全过程”的交互逻辑、画笔库工具系统与架构、“创作 - 管理 - 分享”的闭环体验，实现三大核心价值：一是降低油画创作门槛，利用 AI 灵感库让零基础用户快速掌握基础笔触与色彩搭配；二是提升沉浸式绘画体验，设计丰富画布库与画笔库系统，并配合 AI 辅助油画创作，简化创作步骤；三是构建“创作 - 交流 - 素养沉淀”的社区生态界面设计，助力用户在实践完成后通过分享理解油画美学与技法知识、获得绘画社交的正向情感反馈。最终通过此交互设计研究作为典型案例，填补“全链路 AI + 数字油画 + 社区交互”融合的产品交互设计研究领域的空白。

2.2 主要研究内容

研究内容将围绕 APP 四大核心界面与全链路交互逻辑展开

(1) APP 界面设计与原型设计

包含 AI 灵感发散界面设计、画板界面的 AIGC 辅助交互、画笔种类选择、画笔参数调节面板（厚度、透明度、肌理强度）与画布选择模块（粗纹亚麻布、细纹棉布等）以及工具系统价格设计、画廊管理界面设计，可实现“作品分类存储（按风格 / 创作时间）、AI 优化版本回溯”的交互设计逻辑与原型设计、社区图库界面设计、标签化作品检索（按大师风格 / 画布类型 / 画笔类型）、创作故事联动（作品关联灵感来源与笔记）、油画交流评论区”等界面交互设计。

(2) UI 视觉规范与交互设计规范

使用 Figma 完成从低保真原型（梳理四大界面的跳转逻辑与核心操作流程）到高保真原型（设计界面布局、字体配色规范、交互动效）的界面设计。

(3) 用户测试与体验验证

设计对照实验（实验组使用全链路 APP，对照组使用传统单一功能数字油画工具），验证交互体验指标（绘画创作过程体验满意度、全链路操作流畅性评分），通过问卷与访谈收集反馈，迭代优化 APP 交互设计。

2.3 拟解决关键问题

全链路 AIGC 与数字油画绘画的交互设计研究空白问题、现实油画画笔种类与画布种类库数据转入数字油画 app 的兼容性与真实性和油画创作的流畅度用户体验、数字油画绘画社区生态与创作流程的深度衔接缺陷、AIGC 辅助与用户创作主导权在交互设计中的平衡性。

2. Research Objectives, Main Research Content, and Key Issues to be Solved

2.1 Research Objectives

This project targets oil painting creators of all ages. It addresses core issues found in current digital oil painting applications, such as the fragmentation of the comprehensive workflow (the disconnection among inspiration, creation, management, and sharing), the scarcity and lack of systematic organization in canvas and brush libraries, the fragmentation of AIGC assistance, and the segregation of the community ecosystem from the creative process. The objective is to design an interaction system for a digital oil painting application that integrates AIGC across all four stages of the painting process: pre-painting, during painting, post-painting, and sharing. The design includes features such as an AI inspiration generation interface, a drawing board interface, canvas and brush libraries, a gallery management interface, and a community library.

By constructing interaction logic where AIGC deeply collaborates throughout the entire painting process, developing a systematic brush tool architecture, and creating a closed-loop experience of "creation, management, and sharing," this project aims to achieve three core values. First, it lowers the barrier to oil painting creation by using the AI inspiration library to help users with no prior basis quickly master basic brushstrokes and color schemes. Second, it enhances the immersive painting experience by designing rich canvas and brush systems combined with AI assistance to simplify creative steps. Third, it builds a community ecosystem designed for "creation, communication, and knowledge accumulation," which helps users understand oil painting aesthetics and techniques through sharing and receiving positive emotional feedback after completing their practice. Ultimately, this interaction design research serves as a typical case study to fill the gap in product interaction design research regarding the integration of "full-process AI, digital oil painting, and community interaction."

2.2 Main Research Content

The research content will focus on the four core interfaces of the application and the interaction logic of the comprehensive workflow.

(1) App Interface Design and Prototype Design

This section covers the design of the AI inspiration generation interface, AIGC-assisted interaction on the drawing board, brush type selection, brush parameter adjustment panels (including thickness, opacity, and texture intensity), and canvas selection modules (such as coarse texture linen and fine texture cotton). It also includes the design of the tool system architecture and the gallery management interface, which enables interaction logic and prototyping for classifying works by style or creation date and backtracking AI-optimized versions. Furthermore, it involves the design of the community library interface, tag-based work retrieval (by master style, canvas type, or brush type), creative story linkage

(associating works with inspiration sources and notes), and the oil painting comment section.

(2) UI Visual Standards and Interaction Design Standards

This involves using Figma to complete the interface design, progressing from low-fidelity prototypes that outline the transition logic and core operation flows of the four major interfaces to high-fidelity prototypes. The high-fidelity design includes interface layouts, font and color scheme standards, and interactive motion effects.

(3) User Testing and Experience Verification

This section involves designing control experiments where the experimental group uses the full-process application and the control group uses traditional single-function digital oil painting tools. The goal is to verify interaction experience metrics, such as satisfaction with the painting creation process and the fluency score of full-process operations. Feedback will be collected through questionnaires and interviews to iteratively optimize the interaction design of the application.

2.3 Key Issues to be Solved

The research aims to address several critical issues: the lack of research on interaction design combining full-process AIGC with digital oil painting; the compatibility and realism of transferring physical oil brush and canvas data into a digital application environment, along with the user experience regarding the fluency of oil painting creation; the deep structural defects connecting the digital painting community ecosystem with the creative workflow; and the balance between AIGC assistance and user creative agency within the interaction design.

3、拟采取的研究方法、研究方案及其可行性分析。 **Research methods and research scheme to be adopted and feasibility analysis.**

3.1 拟采取的研究方法

3.1.1 文献研究法

目的是梳理全链路交互设计、AIGC 油画应用、数字艺术社区生态的研究现状，为课题提供理论依据与技术参考。通过知网、谷歌学术网重点分析国内外全链路功能设计缺陷（如灵感 - 创作脱节）、油画画笔与画布种类数据库、社区与创作融合的典型案列，提炼可复用的交互设计框架与交互原则，形成文献综述报告。

3.1.2 用户中心设计法（UCD）

目的是围绕“AI 灵感发散、画板、画廊管理、社区图库”四大界面，设计符合初学者需求的交互逻辑。

3.1.3 对照实验与数据分析法

目的是验证全链路 APP 的交互体验与教育价值。设计对照实验（样本量 50 人左右，随机分为实验组与对照组）：①实验组使用本课题设计的全链路 APP，可通过 AI 灵感辅助创作、画廊管理作品、社区交流绘画体验；②对照组使用传统数字油画工具（如 procreate 基础版，无 AIGC 与社区功能）。实验任务为“创作一幅印象派风格油画”，收集两类数据：①客观指标（创作耗时、画笔选择次数、画布选择次数、社区互动频次）；②主观指标（用户满意度问卷，含“笔触真实度”“操作流畅性”“知识获取感”3 个维度，采用 5 分制）。通过 SPSS 进行独立样本 T 检验，分析两组数据差异，验证 APP 的有效性。

3.2 研究方案

本课题按“前期准备 - 设计开发 - 测试优化 - 总结输出”四阶段推进，周期预计 12 个月，具体任务为前期准备阶段，完成文献综述，明确研究空白，开展用户调研（访谈 + 问卷），确定需求优先级。并完成 AIGC 技术选型与交互框架设计。在交互设计阶段，完成四大界面低保真→高保真原型迭代并开发 APP demo（使用 figma 的 flow 功能绘制动态交互原型界面设计），实现核心功能的交互界面设计：AI 灵感生成、油画画笔与画布种类库选择、画廊管理、社区分享。最后构建油画大师技法知识库的交互界面设计，包含社区评论区技法交流内容。然后是测试优化阶

段，开展对照实验，收集用户数据，分析实验结果，定位问题（如 AIGC 体验问题、社区互动率设计问题）然后迭代优化 APP 交互界面设计。最后的总结输出阶段，整理研究数据与设计文档，撰写设计说明书，提炼 “全链路 AIGC + 数字油画” 交互设计范式，形成 APP 设计规范（含界面组件库、交互逻辑图，输出为 Figma 链接与运行文件）。

3.3 可行性分析

技术可行性方面，现有技术已能支撑核心需求。AIGC 层面，Stable Diffusion 开源模型支持文本 / 草图生成油画风格图像，ComfyUI 提供可视化工作流适配，可快速集成油画专用节点；交互开发层面，Figma 平台框架支持 web 端成熟的界面设计工具，无需从零构建技术体系。

用户可行性方面，目标用户（全年龄段油画初学者）需求明确且市场容量大，据 TalkingData 2024 年报告，国内数字油画 APP 用户中 72% 为零基础初学者，83% 存在 “灵感不足” “油画笔库与画布库难掌握” 的痛点，本课题设计的 “全链路 AIGC 辅助 + 社区交流” 功能可精准匹配需求。

3. Proposed Research Methods, Research Scheme, and Feasibility Analysis

3.1 Proposed Research Methods

3.1.1 Literature Research Method

The purpose of this method is to review the current research status of full-process interaction design, AIGC oil painting applications, and digital art community ecosystems to provide a theoretical basis and technical reference for the project. By using databases such as CNKI and Google Scholar, the study will focus on analyzing domestic and international defects in full-process functional design, such as the disconnection between inspiration and creation. It will also examine databases for oil brushes and canvas types, as well as typical cases of integrating community with creation. The goal is to extract reusable interaction design frameworks and principles to form a literature review report.

3.1.2 User-Centered Design (UCD)

The purpose of this method is to design interaction logic that meets the needs of beginners, centering on the four major interfaces: AI inspiration generation, drawing board, gallery management, and community library.

3.1.3 Controlled Experiment and Data Analysis

The purpose of this method is to verify the interaction experience and educational value of the full-process application. A controlled experiment will be designed with a sample size of approximately 50 participants, randomly divided into an experimental group and a control group. The experimental group will use the full-process application designed in this project, which allows for AI-assisted creation, gallery management, and community exchange of painting experiences. The control group will use traditional digital oil painting tools, such as the basic version of Procreate, without AIGC and community functions. The experimental task is to create an Impressionist-style oil painting. Two types of data will be collected: objective indicators, including creation time, number of brush selections, number of canvas selections, and frequency of community interactions; and subjective indicators derived from a user satisfaction questionnaire covering three dimensions: brush realism, operational fluency, and sense of knowledge acquisition, using a 5-point scale. An independent samples t-test will be performed using SPSS to analyze the differences between the two groups and verify the effectiveness of the application.

3.2 Research Scheme

This project will proceed in four stages: preliminary preparation, design and development, testing and optimization, and summary and output. The expected cycle is 12 months.

The specific tasks for the preliminary preparation stage include completing the literature

review, clarifying research gaps, conducting user research through interviews and questionnaires, and determining priority requirements. AIGC technology selection and interaction framework design will also be completed.

In the design and development stage, the iteration from low-fidelity to high-fidelity prototypes for the four major interfaces will be completed, and an application demo will be developed using the flow function in Figma to create dynamic interactive prototype designs. This includes the interaction interface design for core functions: AI inspiration generation, oil brush and canvas type library selection, gallery management, and community sharing. Finally, the interaction interface for the oil painting master technique knowledge base will be constructed, including technique exchange content in the community comment section.

The next stage is testing and optimization, which involves conducting controlled experiments, collecting user data, analyzing experimental results, identifying problems such as AIGC experience issues or community interaction rate design issues, and then iteratively optimizing the application interaction interface design.

In the final summary and output stage, research data and design documents will be organized to write the design specifications. This stage aims to extract the interaction design paradigm of "Full-process AIGC + Digital Oil Painting" and form application design standards, including interface component libraries and interaction logic maps, output as Figma links and executable files.

3.3 Feasibility Analysis

In terms of technical feasibility, existing technologies can support the core requirements. At the AIGC level, the open-source Stable Diffusion model supports generating oil painting style images from text or sketches, and ComfyUI provides a visual workflow adaptation that allows for the rapid integration of nodes dedicated to oil painting. At the interaction development level, the Figma platform framework supports mature web-based interface design tools, eliminating the need to build a technical system from scratch.

Regarding user feasibility, the target users, who are oil painting beginners of all ages, have clear needs and represent a large market capacity. According to a 2024 report by TalkingData, 72% of domestic digital oil painting application users are zero-basis beginners, and 83% experience pain points such as lack of inspiration and difficulty in mastering oil brush and canvas libraries. The "Full-process AIGC Assistance + Community Exchange" function designed in this project can precisely match these needs.

4、课题的创新点 Novelties of the proposed topic.

本课题以“AIGC + 油画画笔与画布种类数据库 + 全链路交互 + 社区生态”为聚焦点，目标在现有人机共创系统、油画数字艺术交互设计以及AIGC 艺术创作研究的基础上，形成具有媒介特定性和流程完整性的创新框架。在生成式 AI 技术迅速渗透艺术创作、数字绘画工具加速迭代的背景下，现有研究普遍存在两类局限：一是以算法或单一功能为中心，缺乏与完整创作流程相结合的交互设计框架；二是多从“通用图像生成”出发，较少深入到油画等具体媒介的技法规律与用户体验需求。人机共创研究强调“流程视角”和“用户控制权”的重要性，却尚缺少面向特定媒介、特定任务场景的系统化落地方案。在此基础上，本课题拟在交互设计与人机交互领域形成以下几个方面的创新。课题的目标创新点体现在以下四个方面。

4.1 研究视角上的创新——从“单点功能”转向“面向油画场景的全链路 AIGC 交互设计框架”

已有关于 co-creative 系统与 AI 创造力支持工具的研究，多将 AI 工具视为“某一阶段”的辅助者，例如只用于灵感发散、风格迁移或后期润饰，难以覆盖“灵感—创作—评估—分享”的完整创作流程。本课题将主动把研究对象扩展到全流程层面，以全年龄段油画创作者为目标用户，围绕“绘画前—绘画中—绘画完成后—分享”四个阶段交互界面设计，系统规划——AI 灵感发散界面（支持文本 / 草图 / 参考图驱动的多样化油画灵感生成）、画板界面（将 AIGC 作为绘制过程中的实时辅助者，而非仅在结果端出场）、画廊管理界面（融入 AI 优化版本回溯与过程可视化）、社区图库界面（把创作过程数据与社区交流、技法沉淀打通）。

本课题将据此提出一个面向油画场景的全链路 AIGC 交互设计框架，目标是在理论层面回应现有文献中对“过程中心”“工作流导向”人机共创环境的呼吁，并为后续其他艺术媒介的系统设计提供可迁移的结构范式。

4.2 方法层面的创新——构建丰富且结构化的油画画笔库与画布库

本课题在方法层面的核心创新，将体现在不再把“画笔”和“画布”仅作为界面中的少量预设图标使用，而是面向交互设计系统性构建一个丰富且结构化的油画画笔库与画布库，并将其作为数字油画 APP 工具系统的基础。

课题将依据材料学与教学实践,对画笔按毛料(如猪鬃、貂毛、化纤等)、笔形(圆头、平头、榛形、扇形等)、型号区间(如 0-24 号)、软硬与弹性以及常用技法用途等维度进行编码,对画布按材质基底(亚麻、棉布、棉麻混纺等)、纹理级别(粗纹/中纹/细纹/特殊纹理)、织造方式(平纹/斜纹)、底料类型(丙烯底、油性底、改性底)及适配风格与学习阶段等维度进行编码;并将在文献梳理、市场调研及教师/职业画家访谈的基础上,归纳出现实创作中高频使用的画笔-画布配置,整理为层次清晰的工具分类体系与参数数据表。

在此基础上,课题将在交互界面中设计和实现一套明显区别于现有数字油画 APP 的工具选择与管理机制,即用户将能够按“材质—形状—型号”等专业维度进行精细筛选,也将能够围绕“创作目标/类型标签”(如“厚涂练习”“写实人物”“印象派风景”)调用系统推荐的画笔-画布组合。同时,在画廊与社区界面中,每幅作品将与其使用的画笔和画布配置进行关联展示,形成“作品—工具—类型”的结构化映射,支持按工具类型和类型偏好进行浏览与检索。通过上述工作,课题拟在工具系统层面显著提升数字油画 APP 中画笔库与画布库的“种类丰富度”和“交互可见度”,把原本隐性的材料知识转化为用户可直接操作和学习的工具体系,形成本研究在方法层面的关键创新目标。

4.3 交互机制上的创新——分阶段设计“AI 辅助—用户主导”控制模型

近期多项研究已警示,若缺乏合理的交互机制,频繁依赖 AI 工具可能通过“认知卸载”削弱用户的批判性思维与深度加工能力,与此同时,co-creative 系统的评述与框架普遍将“用户控制权、AI 主动性、介入节奏”视为需要精细设计的关键维度。

本课题将结合油画创作的阶段特征,在交互设计中系统规划 AI 介入强度与用户主导权的分配机制,在绘画前阶段,将 AIGC 设计为由用户驱动灵感生成器,通过参数化提示与多方案推荐强化用户对主题、风格与构图的决策权;在绘画中阶段,将 AIGC 作为响应式助手,根据当前笔触与画面状态提供局部笔触生成、色彩校正等可选建议,所有修改均需由用户显式确认,以避免 AI 直接接管画面;在绘画完成后阶段,则主要用于作品诊断、优化版本生成与版式推荐,并通过版本回溯与并列对比界面,确保用户能够回到任一历史阶段、保持对创作演化路径的掌控。通过构建这一分阶段、可调节的控制模型,课题将探索如何在利用 AIGC 提升创造力的同时不削弱学习者主体性,为后续人机共创系统提供具备具体情境细节的设计参考。

4. Novelties of the Proposed Topic

This project focuses on the integration of "AIGC, an Oil Brush and Canvas Type Database, Full-Link Interaction, and Community Ecosystem." It aims to form an innovative framework characterized by medium specificity and process integrity, building upon existing research in human-machine co-creation systems, digital art interaction design for oil painting, and AIGC art creation. In the context where generative AI technology is rapidly penetrating art creation and digital painting tools are accelerating their iteration, existing research generally presents two types of limitations. First, studies often center on algorithms or single functions, lacking an interaction design framework combined with a complete creative workflow. Second, research frequently starts from "general image generation," rarely delving into the technical laws and user experience requirements of specific media such as oil painting. Although research on human-machine co-creation emphasizes the importance of a "process perspective" and "user control," there is still a lack of systematic implementation schemes for specific media and specific task scenarios. Based on this, the proposed project intends to form innovations in the fields of interaction design and human-computer interaction in the following aspects. The targeted innovations of the project are reflected in the following four areas.

4.1 Innovation in Research Perspective: Shifting from "Single-Point Function" to "Full-Link AIGC Interaction Design Framework Oriented to Oil Painting Scenarios"

Existing research on co-creative systems and AI creativity support tools mostly views AI tools as assistants in a "certain stage," such as being used only for inspiration divergence, style transfer, or post-processing retouching. Consequently, it is difficult to cover the complete creative workflow of "inspiration, creation, evaluation, and sharing." This project will actively extend the research object to the full process level, targeting oil painting creators of all ages. It involves the systematic planning of interaction interface designs around four stages: pre-painting, during painting, post-painting, and sharing. This includes the AI inspiration generation interface, which supports diverse oil painting inspiration generation driven by text, sketches, or reference images; the drawing board interface, which positions AIGC as a real-time assistant during the drawing process rather than merely appearing at the result end; the gallery management interface, which incorporates backtracking of AI-optimized versions and process visualization; and the community library interface, which connects creation process data with community communication and the accumulation of techniques.

Based on this, the project will propose a full-link AIGC interaction design framework oriented toward oil painting scenarios. The goal is to theoretically respond to the calls in existing literature for a "process-centric" and "workflow-oriented" human-machine co-creation environment, and to provide a transferable structural paradigm for the system design of other artistic media in the future.

4.2 Innovation in Methodology: Constructing a Rich and Structured Oil Brush and Canvas Library

The core innovation of this project at the methodological level lies in no longer treating "brushes" and "canvases" merely as a few preset icons in the interface. Instead, it involves systematically constructing a rich and structured oil brush and canvas library oriented toward interaction design, serving as the foundation of the digital oil painting app tool system.

Drawing on materials science and teaching practices, the project will encode brushes based on dimensions such as bristle material (hog bristle, sable, synthetic fiber), brush shape (round, flat, filbert, fan), size range (such as No. 0 to 24), softness, elasticity, and common technical uses. Similarly, canvases will be encoded based on dimensions such as material base (linen, cotton, cotton-linen blend), texture level (coarse, medium, fine, special texture), weaving method (plain, twill), primer type (acrylic base, oil base, modified base), and adaptation to styles and learning stages. Based on literature reviews, market research, and interviews with teachers and professional painters, the project will summarize the brush and canvas configurations frequently used in realistic creation, organizing them into a clearly layered tool classification system and parameter data table.

On this basis, the project will design and implement a tool selection and management mechanism in the interaction interface that is distinctly different from existing digital oil painting apps. Users will be able to perform fine screening according to professional dimensions like "material, shape, and model," and will also be able to call upon system-recommended brush and canvas combinations centered on "creative goals or type tags" (such as "impasto practice," "realistic character," or "Impressionist landscape"). Simultaneously, in the gallery and community interfaces, each work will be displayed in association with its used brush and canvas configuration, forming a structured mapping of "Work, Tool, and Type." This supports browsing and retrieval based on tool type and preference. Through this work, the project aims to significantly improve the "variety richness" and "interaction visibility" of brush and canvas libraries in digital oil painting apps at the tool system level, transforming originally implicit material knowledge into a tool system that users can directly operate and learn from. This constitutes a key innovation goal of this research at the methodological level.

4.3 Innovation in Interaction Mechanism: Phased Design of the "AI Assistance–User Dominance" Control Model

Recent studies have warned that without reasonable interaction mechanisms, frequent reliance on AI tools may weaken users' critical thinking and deep processing abilities through "cognitive offloading." Meanwhile, reviews and frameworks of co-creative systems universally regard "user control, AI initiative, and intervention rhythm" as key dimensions requiring refined design.

This project will combine the stage characteristics of oil painting creation to

systematically plan the distribution mechanism of AI intervention intensity and user dominance within the interaction design. In the pre-painting stage, AIGC is designed as a user-driven inspiration generator, strengthening user decision-making power over themes, styles, and composition through parameterized prompting and multi-scheme recommendations. In the painting stage, AIGC acts as a responsive assistant, providing optional suggestions such as local brushstroke generation and color correction based on current brushstrokes and image status; importantly, all modifications must be explicitly confirmed by the user to avoid AI directly taking over the image. In the post-painting stage, AIGC is primarily used for work diagnosis, optimized version generation, and layout recommendation. Through version backtracking and side-by-side comparison interfaces, the system ensures users can return to any historical stage and maintain control over the evolutionary path of creation. By constructing this phased and adjustable control model, the project will explore how to use AIGC to enhance creativity without weakening learner agency, providing design references with specific contextual details for subsequent human-machine co-creation systems.

5、计划进度、预期成果 Research schedule, and expected outcomes

时间阶段 (2025.12 – 2027.01)	研究任务	预期成果
2025 年 12 月 – 2026 年 2 月 (前期准备阶段)	① 开展系统文献综述：梳理 AIGC、人机共创、数字油画 APP、油画笔库与画布库、艺术社区交互等相关研究； ② 明确研究空白与研究问题，初步搭建“全链路 AIGC + 数字油画 + 社区交互”的理论框架； ③ 设计并实施用户调研方案（深度访谈 + 问卷调查），覆盖零基础用户、爱好者与专业学习者； ④ 基于调研结果整理用户需求与痛点，形成需求优先级矩阵； ⑤ 开展 AIGC 技术选型（模型、接口、能力范围）与可行性分析，初步构思全链路交互框架与信息架构。	· 文献综述与研究现状分析报告； · 研究问题与研究目标的理论框架草案； · 用户调研记录（访谈录音/访谈纪要、问卷数据）及分析报告； · AIGC 技术选型与系统架构初步方案； · 全链路交互结构草图与流程图（低保真）。
2026 年 3 月 – 2026 年 8 月 (设计阶段)	① 基于前期框架，进行四大核心界面（AI 灵感发散界面、画板界面、画廊管理界面、油画笔库与画布库界面、社区图库界面）的信息架构与低保真原型设计； ② 在 Figma 中完成从低保真到中保真原型的迭代，理清四大界面之间的跳转逻辑与核心操作流	· 四大核心界面的 IA (信息架构图) 与用户流程图； · 低保真 → 中保真 → 高保真界面原型文档； · Figma 交互原型 (含完整 flow 的 APP demo, 可点击演示)； · 关键功能交互方案说明书 (AI 灵感生成、笔触模

	程 (flow) ； ③ 重点设计 “AI 灵感生成”、 “油画笔库与画布库（粗纹亚麻布、细纹棉布等）”、“作品分类存储与 AI 优化版本回溯”、“社区标签化检索与油画评论区”等关键交互界面； ④设计并搭建“油画笔库与画布库”的信息结构与界面原型，将作品、类型标签与社区讨论关联； ⑤ 在 Figma 中制作可操作的高保真动态交互原型 (APP demo) ，形成初步可演示的全链路体验。	拟、画廊管理、社区分享与油画交流等) ； · 油画笔库与画布库知识库的结构设计与界面原型； · 初步设计规范（颜色、字体、组件基础样式）。
2026 年 9 月 – 2026 年 11 月 (测试优化阶段)	①设计对照实验方案：实验组使用全链路 AIGC 数字油画 APP 原型，对照组使用传统单一功能数字油画工具（或现有商用 APP）； ② 招募不同水平用户（零基础、爱好者、艺术类学生等），组织可用性测试与任务完成实验，记录操作过程与过程反馈；③ 量化与质性并行收集数据（任务完成时间、操作出错率、主观满意度、笔触真实度感知、全链路流畅性评分、社区互动意愿等）； ④ 使用统计与质性分析方法对实验数据进行分析，对比两类工具在创作门槛、沉浸感与操作负	· 对照实验设计方案与伦理/招募说明； · 用户测试原始数据（录屏、观察记录、问卷量表、访谈纪要等）； · 数据分析报告（包括定量统计与定性编码结果）； · 问题清单与优先级（AIGC 体验、笔触真实度、全链路流畅性、社区参与度等）； · 优化后的高保真交互原型（第二版/第三版）； · 阶段性小结报告，为论文实证章节提供数据基础。

	担等维度的差异； ⑤ 基于结果定位主要问题（如 AIGC 介入时机不当、笔触反馈不足、社区互动入口隐蔽等），提出针对性的交互优化策略，并对 Figma 原型进行一轮或多轮迭代。	
2026 年 12 月 - 2027 年 1 月（总结输出阶段）	①综合前期文献研究、设计过程与实验结果，系统整理“全链路 AIGC + 数字油画 + 油画笔库与画布库 + 社区交互”的设计理论与方法论总结； ②完成 APP 设计规范输出，包括界面组件库、图标与控件样式、布局与栅格规范、交互反馈模式及状态图等； ③将交互逻辑图、用户流程图、信息架构与关键界面说明整理为完整的设计文档，并以 Figma 链接与可运行文件形式归档； ④撰写硕士学术作品类论文，完成绪论、相关研究、方法与设计过程、实验与分析、结论与展望等章节的撰写与修改； ⑤ 准备开题/中期/答辩所需展示材料（包括演示视频、关键界面展示与研究成果可视化），完成学位论文提交与作品归档。	· “全链路 AIGC + 数字油画”交互设计范式的理论总结与框架图； · 完整 APP 设计规范（UI 视觉规范、组件库、交互模式说明等）； · Figma 最终版原型链接与打包运行文件（可作为作品成果提交）； · 研究过程与设计成果的综合报告及演示文档； · 硕士学术作品类论文定稿； · 供后续研究或产业实践参考的设计规范与案例。

Time Frame	Research Tasks	Expected Outcomes
Dec 2025 – Feb 2026 (Preliminary Preparation Phase)	1. Conduct a systematic literature review: Cover areas such as AIGC, human-computer co-creation, digital oil painting applications, oil brush and canvas libraries, and art community interaction. 2. Identify research gaps and questions: Draft the preliminary theoretical framework for "Full-Link AIGC + Digital Oil Painting + Community Interaction." 3. Design and implement user research: Conduct in-depth interviews and questionnaires covering zero-basis users, enthusiasts, and professional learners. 4. Analyze user needs: Organize user requirements and pain points based on survey results to form a requirement priority matrix. 5. Conduct AIGC technology selection: Analyze models, APIs, and capabilities, perform feasibility analysis, and conceptualize the initial full-link interaction framework and information architecture.	• Literature review and status analysis report. • Draft theoretical framework regarding research questions and objectives. • User research records (interview transcripts, questionnaire data) and analysis reports. • Preliminary AIGC technology selection and system architecture plan. • Low-fidelity sketches and flowcharts of the full-link interaction structure.
Mar 2026 – Aug 2026 (Design Phase)	1. Develop Information Architecture (IA) and prototypes: Design low-fidelity prototypes for the four core interfaces (AI inspiration generation, drawing board, gallery management, brush and canvas libraries, and community library) based on the preliminary	• Information Architecture (IA) maps and user flowcharts for the four core interfaces. • Low-fidelity to high-fidelity interface prototype documentation. • Figma interactive prototype (clickable APP

	framework. 2. Iterate prototypes in Figma: Progress from low-fidelity to mid-fidelity prototypes, clarifying transition logic and core operation flows between the four major interfaces. 3. Design key interaction interfaces: Focus on "AI Inspiration Generation," "Oil Brush and Canvas Libraries (e.g., coarse linen, fine cotton)," "Work Classification Storage and AI Optimized Version Backtracking," and "Community Tagged Retrieval and Oil Painting Comment Section." 4. Build library structure: Design the information structure and interface prototypes for the "Oil Brush and Canvas Library," linking artworks and type tags with community discussions. 5. Create high-fidelity dynamic prototypes: Produce an interactive APP demo in Figma to form a preliminary demonstrable full-link experience.	demo with complete flows). • Specification manuals for key function interactions (AI inspiration, brushstroke simulation, gallery management, community sharing, etc.). • Structural design and interface prototypes for the oil brush and canvas knowledge base. • Preliminary design standards (colors, typography, basic component styles).
Sep 2026 – Nov 2026 (Testing and Optimization Phase)	1. Design controlled experiments: The experimental group uses the full-link AIGC digital oil painting APP prototype, while the control group uses traditional single-function digital oil painting tools (or existing commercial APPs). 2. Recruit users and test: Recruit users of different skill levels (beginners,	• Controlled experiment design plan and ethics/recruitment statements. • Raw user test data (screen recordings, observation notes, questionnaire scales, interview transcripts). • Data analysis report (including quantitative statistics and qualitative

	enthusiasts, art students) to organize usability testing and task completion experiments, recording the operation process and feedback. 3. Collect data: Gather quantitative and qualitative data (task completion time, error rate, subjective satisfaction, perceived brushstroke realism, full-link fluency score, willingness for community interaction). 4. Analyze data: Use statistical and qualitative methods to compare differences between the two tools in dimensions such as creation barrier, immersion, and operational burden. 5. Optimize design: Identify major issues based on results (e.g., improper AIGC intervention timing, insufficient brush feedback, hidden community interaction entry points), propose targeted interaction optimization strategies, and perform iterations on the Figma prototype.	coding results). • Problem list with priorities (AIGC experience, brush realism, full-link fluency, community engagement, etc.). • Optimized high-fidelity interaction prototype (Version 2 or 3). • Interim summary report providing a data foundation for the empirical chapters of the thesis.
Dec 2026 – Jan 2027 (Summary and Output Phase)	1. Synthesize findings: Systematically summarize the design theory and methodology of "Full-Link AIGC + Digital Oil Painting + Brush/Canvas Library + Community Interaction" based on literature, design processes, and experimental results. 2. Output design specifications: Finalize APP design standards, including interface	• Theoretical summary and framework map of the "Full-Link AIGC + Digital Oil Painting" interaction design paradigm. • Complete APP design specifications (UI visual standards, component libraries, interaction mode descriptions, etc.). • Final Figma prototype link and packaged executable file (submitted as project

	component libraries, icon and control styles, layout and grid standards, interaction feedback modes, and state diagrams. 3. Archive documentation: Compile interaction logic maps, user flowcharts, information architecture, and key interface descriptions into complete design documentation, archived as Figma links and executable files. 4. Write thesis: Complete the master's academic work thesis, including chapters on introduction, related research, methods and design process, experiment and analysis, and conclusions. 5. Prepare for defense: Prepare materials for the proposal, mid-term, and defense (including demo videos, key interface displays, and research visualization), and complete thesis submission and work archiving.	output). • Comprehensive report and presentation documents of the research process and design results. • Final draft of the master's academic work thesis. • Design specifications and cases for reference in subsequent research or industrial practice.
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6、与本课题有关的工作积累、已有的研究工作成绩。 **Prior experience and accomplished achievements related to the proposed topic.**

- (1) 已完成 初版本的 APP 的低保真原型与高保真原型设计稿
- (2) 已开发基于 Unity 的油画绘画 APP demo，实现基础的 “草图绘制 - 简单风格生成” 功能，支持 iOS/Android 端运行。
- (3) 参与相关课题项目《基于物理仿真的教育向 3D 智能油画引擎与交互系统》在 2025 年国际工业博览会上展出，获行业专家 “技术与教育融合性强” 的评价，验证了项目的实践价值。
- (4) 一项国家自然基金课题项目

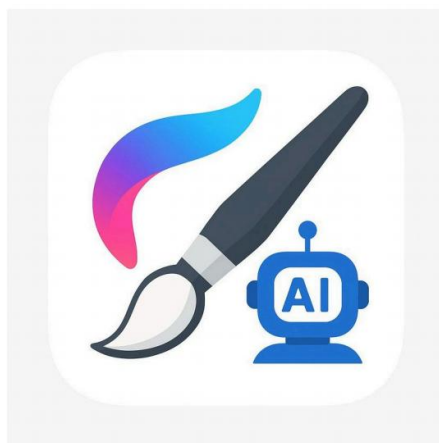


图 1 AIGC 油画 app logo 设计（作者自绘）

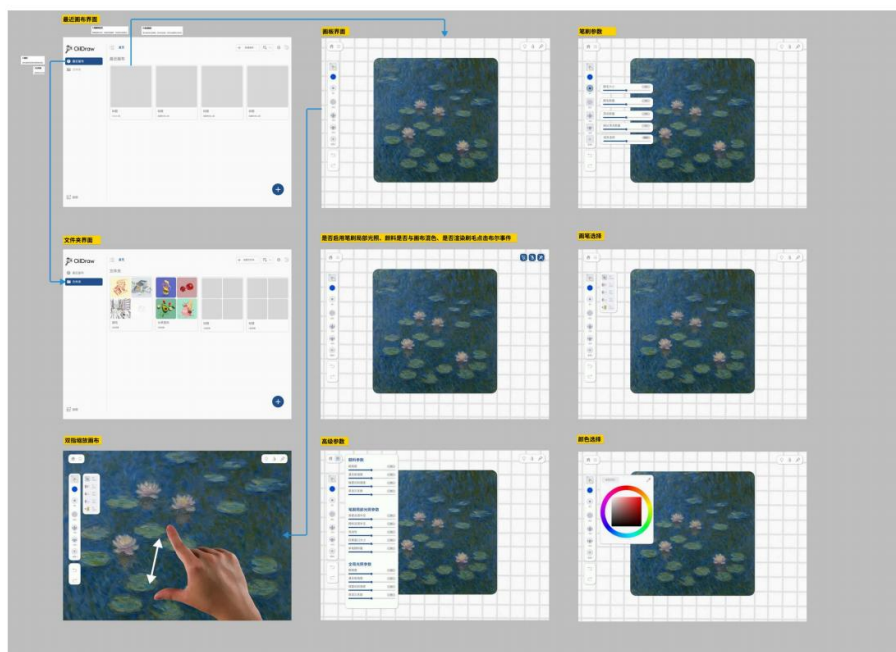


图 2 AIGC 油画 app 交互原型设计初稿（作者自绘）

- (1) The low-fidelity prototype and high-fidelity prototype design drafts of the initial version of the APP have been completed.
- (2) A Unity-based oil painting APP demo has been developed, which implements basic functions of "sketch drawing - simple style generation" and supports operation on iOS/Android platforms.
- (3) The team participated in the exhibition of the related project Physics Simulation-Based Educational 3D Intelligent Oil Painting Engine and Interactive System at the 2025 International Industry Expo. The project received the evaluation of "strong integration of technology and education" from industry experts, verifying its practical value.
- (4) One National Natural Science Foundation project has been involved.

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学生签字/**Signature of Student:** 陈皓宇

导师签字/**Signature of Supervisor** 魏明

日期/**Date:** 2025-12-11